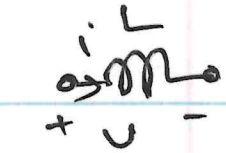
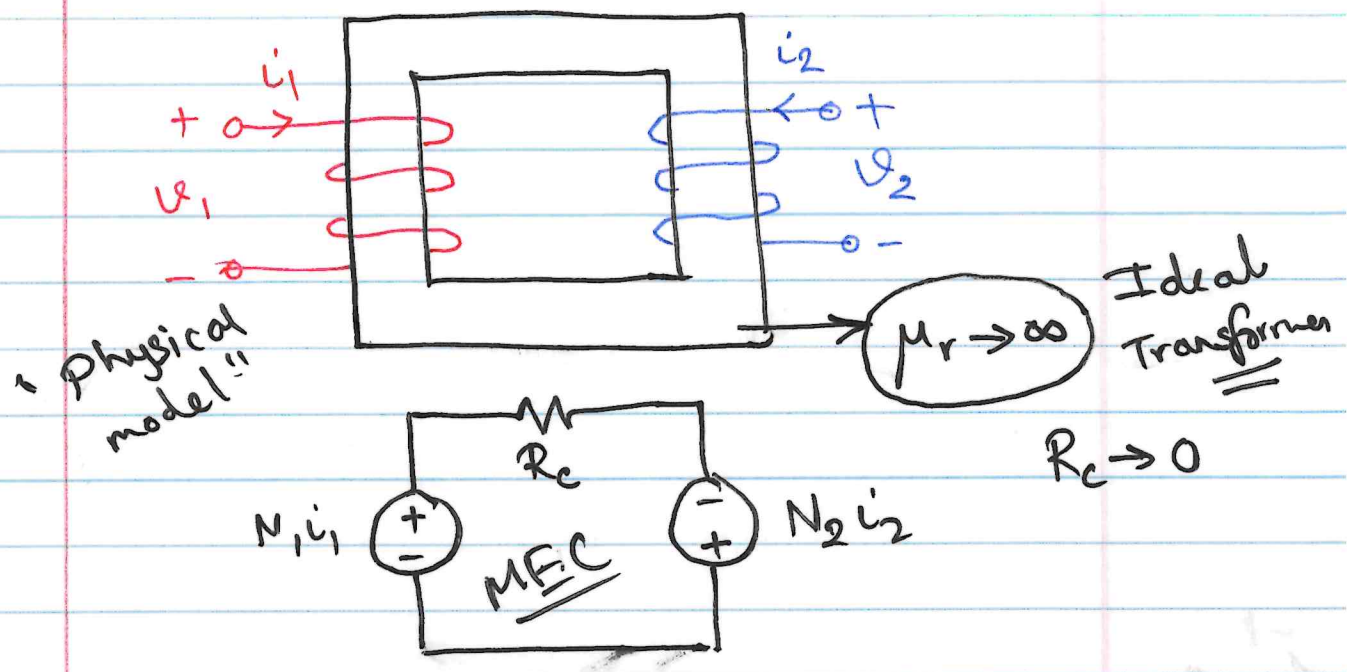
Lecture 9.

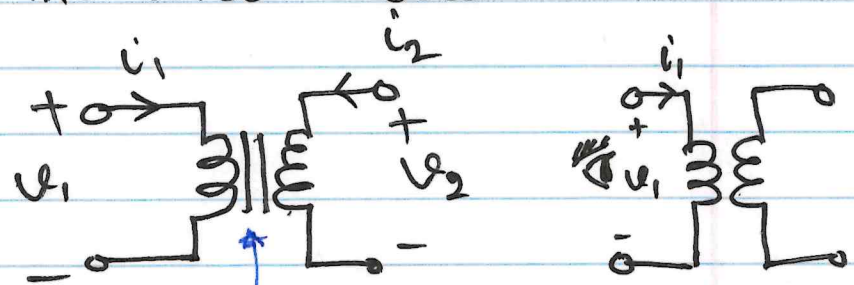




$$\frac{v_1}{N_1} = \frac{v_2}{N_2}$$

$$N_1 i_1 + N_2 i_2 = 0$$

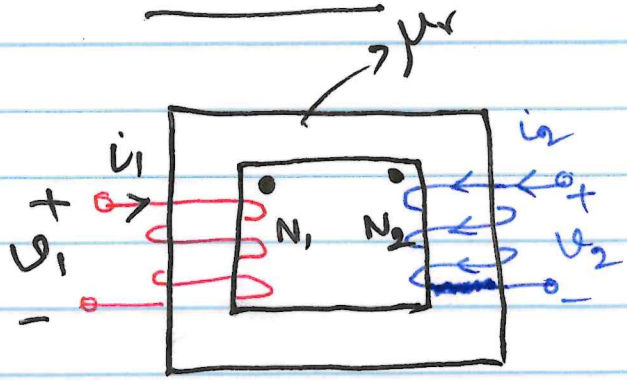
Mathematical model

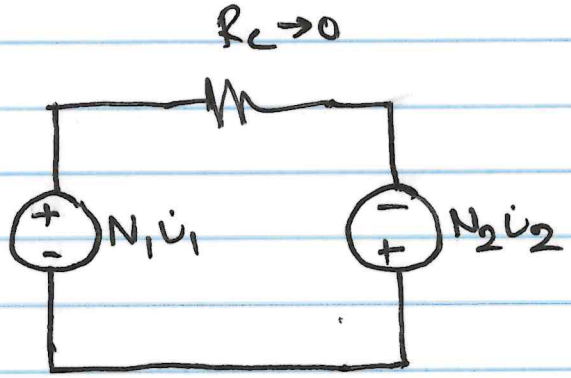


Electrical circuit model

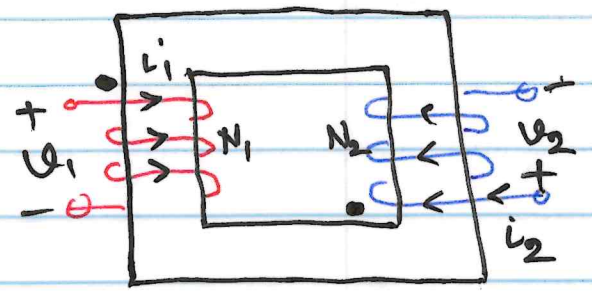
Peter B. Peter G.

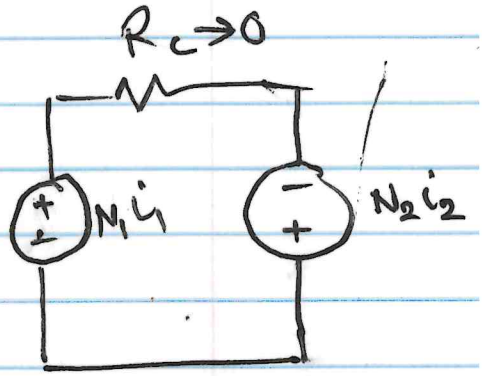
Peter B





Peter G

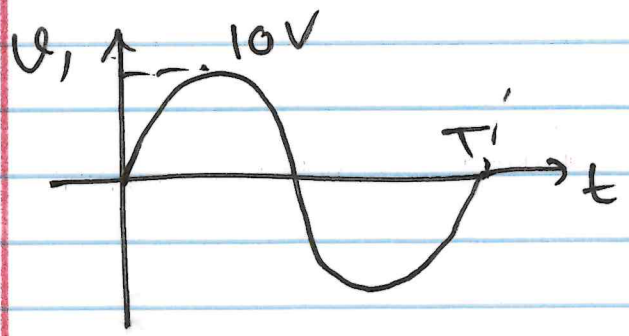


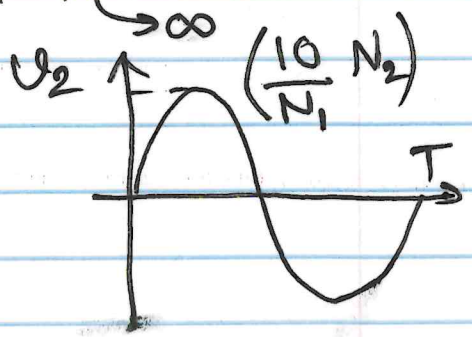


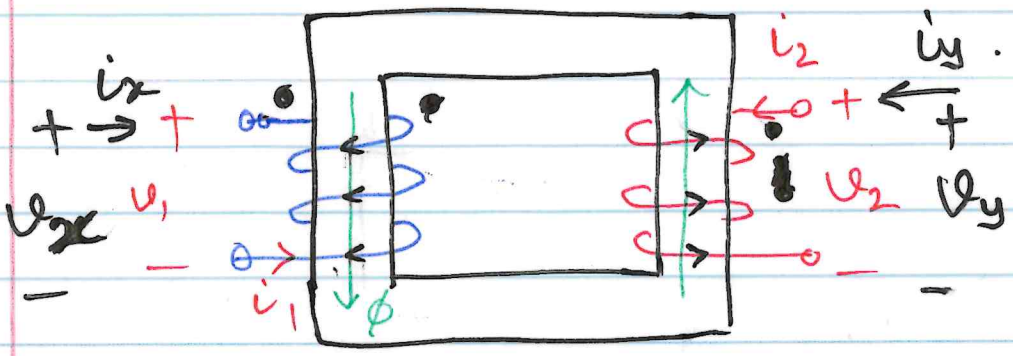
$$N_1 i_1 + N_2 i_2 = 0 \qquad (R_c \to 0,\ \mu_r \to \infty)$$

$$R = \frac{l}{\mu_0 \mu_r A}$$

($\mu_r \to \infty$)

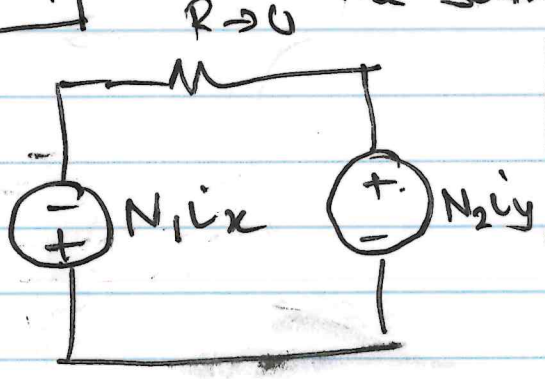






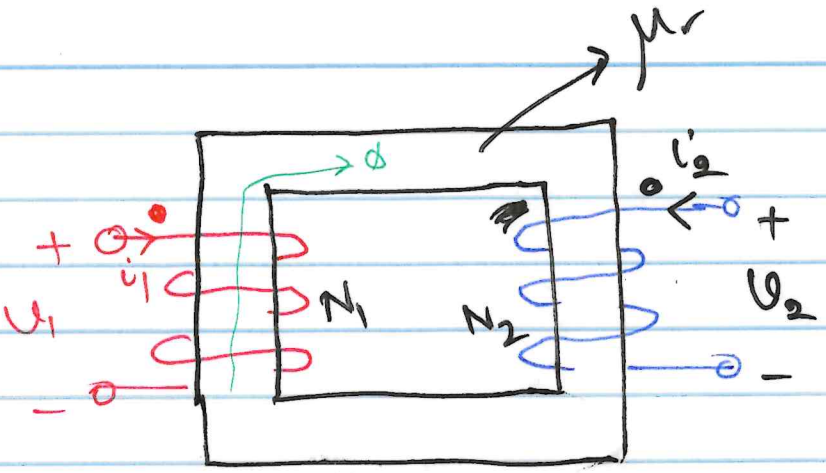
$$v_x = v_1$$
$$i_x = -i_1$$

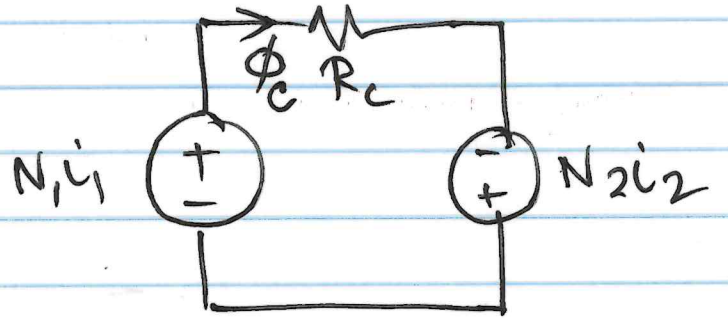
Current entering the dots throws flux in the same direction.

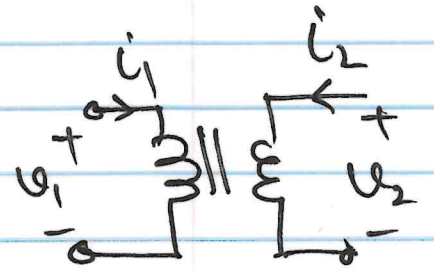


$$v_y = v_2$$
$$i_y = i_2$$

$$N_1 i_x + N_2 i_y = 0$$







$$\frac{v_1}{N_1} = \frac{v_2}{N_2} \quad ; \quad N_1 i_1 + N_2 i_2 = 0$$

when $R_c \to 0$!

μ_r = finite

$$N_1 i_1 + N_2 i_2 = \phi_c R_c$$

Imagine : $i_2 = 0$ [if wdg #2 is OPEN]

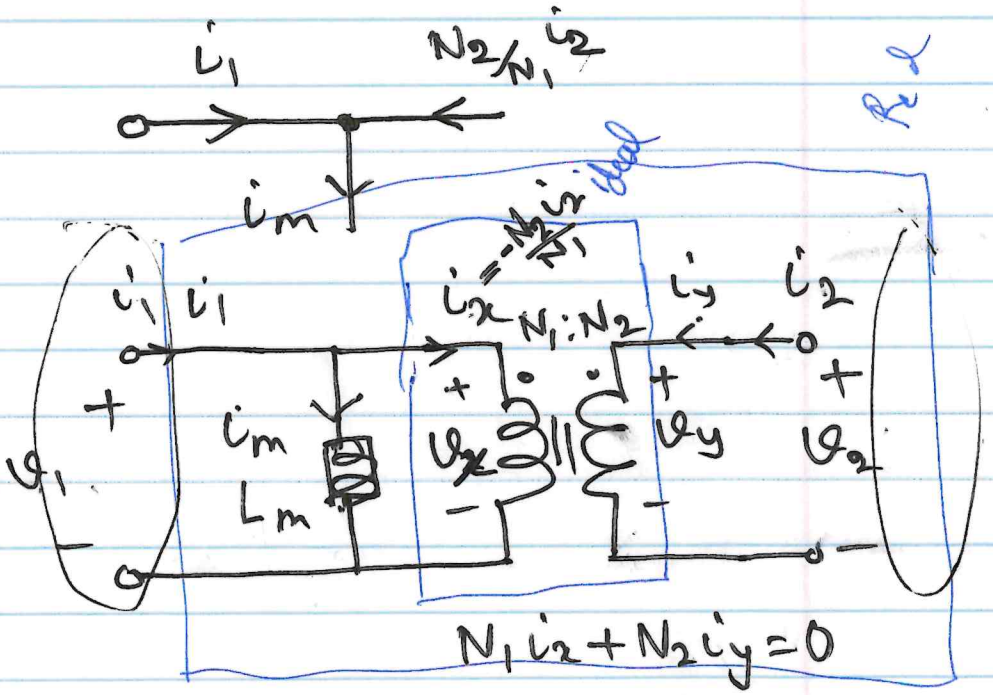
$$i_1 = i_m$$

$$N_1 i_m = \phi_c R_c$$

$$N_1 i_1 + N_2 i_2 = N_1 i_m$$

↑ magnetizing current

$$i_1 + \frac{N_2}{N_1} i_2 = i_m$$



$$\frac{v_x}{N_1} = \frac{v_y}{N_2}$$

$$L_m = \frac{N_1^2}{R_c}$$

magnetizing inductance